



MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL
(A constituent unit of MAHE, Manipal)

COURSE PLAN

Department	Computer Science and Engineering	Programme	B. Tech. (CSE)
Course Name	COMPILER DESIGN	Course Code	CSE 3222
Semester	V	Curriculum	2022
Name of the faculty	Ms. Priya Kamath B.	Academic year	2025-2026
No. of Contact Hours/Week	L T P C: 2 1 0 3		

COURSE OUTCOMES (COs)

At the end of this course, the student should be able to:		No. of Hours	Marks
CO1	Familiarize on different phases of a compiler and recognize steps involved in lexical analyzer generators	6	16
CO2	Understand the top down and bottom up parsing techniques.	12	32
CO3	Identify ambiguous grammars and analyze syntax directed translation techniques.	3	8
CO4	Translate expressions into three address code and elaborate code generation phase	13	34
CO5	Understand the automated tools like FLEX and YACC.	2	5
CO6	Design the grammar and build compiler modules using compiler construction tools	Self-Learning	5
Total		36	100

In semester & End semester plan and schedule (AY: 2026-27)

Component	Type	Max. Marks	Duration	Syllabus: Topics covered during	Schedule	Blooms taxonomy levels
Assignment 1	Quiz	5	20 min	L0 – L10	27 Jan 2026-02 Feb 2026	3 to 6
Assignment 2	Quiz	5	20 min	L10- L18	16 Feb 2026-23 Feb 2026	2 to 6
Mid Sem Exam	Descriptive (E-pad)	30	90 min	L0-L19		2 to 6
Assignment 3	Quiz	5	20 min	L19-L26	23 Mar 2026 - 30 Mar 2026	2 to 6
Assignment 4	Quiz	5	20 min	L27-L36	13 Apr 2026 - 20 Apr 2026	2 to 5
END Semester examination		50	3 Hrs	L1 – L36		2 to 6

LESSON PLAN

Lecture No.	Topic	Course Outcome addressed
L0	Introduction to the course (Introduction between teacher & students. Overview of the subject).	-
L1	Language Processors, The Structure of a Compiler- Lexical Analysis, Syntax Analysis, Semantic Analysis	CO1
L2	Intermediate Code Generation, Code Optimization, Code Generation, Symbol-Table Management	CO1
L3 (T1)	The Role of the Lexical Analyzer, Input Buffering	CO1
L4	Recognition of Tokens, Architecture of a Transition-Diagram-Based Lexical Analyzer	CO1
L5	Design of a Lexical Analyzer Generator- The Structure of the Generated Analyzer, Pattern Matching Based on NFAs.	CO1
L6 (T2)	Problems on Pattern Matching Based on NFAs.	CO1
L7	Syntax Analysis - Introduction, writing a Grammar- Lexical versus Syntactic Analysis	CO2
L8	Eliminating Ambiguity and Left Recursion, Left Factoring	CO2
L9 (T3)	Problems on Eliminating Ambiguity and Left Recursion	CO2
L10	Top-Down Parsing - Recursive-Descent Parsers	CO2
L11	First and Follow, LL(1) Grammars	CO2
L12 (T4)	Examples on Predictive Parsing	CO2
L13	Non-recursive Predictive Parsing	CO2
L14	Error Recovery in Predictive Parsing, Bottom-Up Parsing - Reductions, Handle Pruning	CO2
L15 (T5)	Shift-Reduce Parsing, Introduction to LR parsing- Simple LR, Why LR Parsers?	CO2
L16	Items and LR (0) Automaton, The LR-Parsing Algorithm	CO2
L17	Constructing SLR-Parsing Tables	CO2
L18 (T6)	Examples on LR parsing	CO2
L19	More Powerful LR parsers- Canonical LR (1) Items	CO2
L20	Constructing LR(1) Sets of Items, Canonical LR(1) Parsing Tables	CO2
L21 (T7)	Using Ambiguous Grammars-Precedence and Associativity to Resolve Conflicts	CO3
L22	Syntax-Directed Translation - Syntax-Directed Definitions, Evaluation Order for SDD's- Dependency Graphs	CO3
L23	Ordering the Evaluation of Attributes, Applications of Syntax-Directed Translation -	CO3

L24 (T8)	Construction of Syntax Trees, Intermediate-Code Generation - Variants of Syntax Trees	CO4
L25	Three Address Code- Addresses and Instructions, Quadruples, Triples	CO4
L26	Types and Declarations- Type Expressions, Type Equivalence, Declarations	CO4
L27 (T9)	Problems on Three Address Code	CO4
L28	Translation of Expressions- Operations Within Expressions.	CO4
L29	Tutorial on Translation of Operations within expressions	CO4
L30 (T10)	Code Generation - Issues in Design of Code Generator, The Target Language, Basic Blocks and Flow Graphs	CO4
L31	Optimization of Basic Blocks- The DAG Representation of Basic Blocks	CO4
L32	Tutorial on Optimization of Basic Blocks	CO4
L33 (T11)	Peephole Optimization, Register Allocation and Assignment- Global Register Allocation.	CO4
L34	Tutorial on Peephole Optimization	CO4
L35	Theory of FLEX- Structure of a FLEX program, Regular Expression, FLEX library functions	CO5
L36 (T12)	Theory of YACC- YACC Symbols, Symbol values, Symbol Types, YACC Library	CO5

References:

1	Alfred V. Aho, Monica S. Lam, Ravi Sethi, Jeffrey D. Ullman, "Compilers Principles, Techniques and Tools", Pearson Education, 2nd edition. 2010.
2	Vinu V. Das, "Compiler Design using FLEX and YACC", Prentice-Hall, 2007
3	Kenneth C. Loudon, "Compiler Construction - Principles and Practice", Thomson, First Edition, 2007.
4	John R. Levine, Tony Manson, Doug Brown, "LEX & YACC", O Reilly Media, Second Edition, 2012.

FACULTY MEMBERS TEACHING THE COURSE (IF MULTIPLE SECTIONS EXIST):

FACULTY NAME	SECTION	FACULTY NAME	SECTION
Ms. Priya Kamath B.	A	Ms. Deepthi S.	C
Dr. Raghu T. V	B	Dr. Swathi Prabhu	D

Submitted by: Ms. Priya Kamath B.

(Signature of the faculty)

Date: 09/01/2026

Approved by: Dr. P C Siddhalingaswamy

(Signature of HOD)

Date: 09/01/2026

COURSE PLAN – ADDITIONAL DETAILS

At the end of this course, the student should be able to:		No. of contact Hours	Marks	Program outcomes (POs)	Learning outcomes (LOs)	Program Specific outcomes (PSOs)	Blooms Taxonomy (BT)
CO1	Familiarize on different phases of a compiler and recognize steps involved in lexical analyzer generators	6	16	1,2,8,12	2	2	2 and above
CO2	Understand the top down and bottom up parsing techniques.	14	37	1,8,12	3	2	2 and above
CO3	Identify ambiguous grammars and analyze syntax directed translation techniques.	3	8	1,2,8,12	2	2	3 and above
CO4	Translate expressions into three address code and elaborate code generation phase	11	29	1,2,8,12	5	2	3 and above
CO5	Understand the automated tools like FLEX and YACC.	2	5	1,2,8,12	3	2	2 and above
CO6	Design the grammar and build compiler modules using compiler construction tools	Self Learning	5	5, 9	6, 13	2	4 and above
Total hours/ Marks		36	100				

Mapping of course outcomes (COs) with Program outcomes (POs) and Program Specific outcomes (PSOs)

Course outcomes (COs)		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3	PSO4
CO1	Familiarize on different phases of a compiler and recognize steps involved in lexical analyzer generators	3	1	–	–	–	–	–	1	–	–	–	2	–	2		–
CO2	Understand the top down and bottom up parsing techniques.	2	–	–	–	–	–	–	1	–	–	–	2	–	2		–
CO3	Identify ambiguous grammars and analyze syntax directed translation techniques.	2	3	–	–	–	–	–	1	–	–	–	1	–	2		–
CO4	Translate expressions into three address code and elaborate code generation phase	2	3	–	–	–	–	–	1	–	–	–	2	–	2		–
CO5	Understand the automated tools like FLEX and YACC.	3	3	–	–	–	–	–	1	–	–	–	2	–	2		–
CO6	Design the grammar and build compiler modules using compiler construction tools	–	–	–	–	1	–	–	–	1	–	–	–	–	–	–	–
Average Program Articulation Level		2.4	2.5	0	0	1	0	0	1	1	0	0	1.8	0	2	0	0

MIT/GEN/F-01/R3

Mapping of course learning outcomes (CLOs) with AHEP Learning outcomes (AHEP LOs)

Course Learning Outcomes (CLOs)		C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	C14	C15	C16	C17	C18
CSE 3151.1	Familiarize on different phases of a compiler and recognize steps involved in lexical analyzer generators																		
CSE 3151.2	Understand the top down and bottom up parsing techniques.																		
CSE 3151.3	Identify ambiguous grammars and analyze syntax directed translation techniques.																		
CSE 3151.4	Translate expressions into three address code and elaborate code generation phase																		
CSE 3151.5	Understand the automated tools like FLEX and YACC.																		
CSE 3151.6	Design the grammar and build compiler modules using compiler construction tools																		

Abbreviations

1. CO – Course outcome
2. PO – Program outcome
3. PSO – Program Specific outcome
4. LO – Learning outcome
5. CLO – Course Learning outcome
6. BT – Blooms Taxonomy
7. AHEP – The Accreditation of Higher Education Programmes

